INCH POUND
MIL-D-83934B (USAF)
17 JULY 1989

SUPERSEDING
MIL-D-83834A (USAF)
12 SEPTEMBER 1972

MILITARY SPECIFICATION

DRYER, AIR, LOW PRESSURE

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers air dryers for use in low pressure cable systems. The dryer shall remove particulate matter, oil aerosol, and water vapor from effluent atmospheric air and furnish clean, dry air as specified herein. The unit shall be automatic in operation. It is intended that the parameters given herein be representative of commercially available air dryers.

1.2 Classification.

1.2.1 Types. Air dryers shall be the following types as specified (see 6.2). Inlet air shall be 95° F db and 80° F wb at 14.7 psia pressure.

Type I - Maximum moisture content of effluent air at 14.7 psia and 70° F shall be 310 PPM by weight (equivalent to - 18° F dew point or 2% R.H. at 70° F).

Type II - Maximum moisture content of effluent air at 14.7 psia and 70° F shall be 80 PPM by weight (equivalent to - 40° F dew point).

1.2.2 Capacity. The minimum capacity (effluent air flow rate) expressed as standard cubic feet per day (SCFD) shall be as specified in 6.2.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMIRBC, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 4440

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SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated, Plywood

Military

MIL-P-116	Preservation, Methods of
MIL-D-3716	Desiccants, Activated, for Dynamic Dehumidification
MIL-D-22577	Heating Elements, Electrical, Cartridge, Strip and Tubular Type

STANDARDS

Military

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

National Fire Protection Association

Pamphlet 70	National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Park, Quincy, Mass. 02269.)

American Society of Mechanical Engineers (ASME)

Power Test Codes, Unfired Pressure Vessels Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS.

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2. Design.

3.2.1 Construction. The air dryer shall be a complete unit consisting of a drive motor, compressor, drying system, and other components necessary to provide a fully automatic unit. The unit shall be suitable for continuous or intermittent operation at conditions specified herein. The air dryer shall be housed in a metal enclosure constructed as a single cabinet unit. The maximum size of the unit shall be as specified in 6.2. When specified (see 6.2) the air dryer shall be contained within a weatherproof enclosure suitable for pole mounting. When the dryer design incorporates a combination of refrigerant and desiccant systems, the individual design paragraphs of this specification shall apply.

3.2.2 Design (refrigeration type). When compression-refrigeration units are furnished they shall be the hermetically sealed type and use refrigerant 12 or 22.

3.2.3 Design (desiccant type). When compression desiccant units are furnished, each tower shall have at least one removable head to facilitate replacement of desiccant and allow inspection of tower interior. Desiccant shall conform to the requirements of MIL-D-3716.

3.2.4 Heaters. When heaters are used in the desiccant regenerating cycle, heater elements shall be in accordance with MIL-H-22577.

3.2.5 Storage tank. Air storage tanks or desiccant towers shall be constructed of corrosion resistant material and shall comply with the requirements of ASME code for unfired pressure vessels.

3.2.6 Regulating valve. The air discharge line shall include a pressure regulating valve suitable for setting the outlet pressure at any point between 0 and the maximum design pressure as specified in 6.2.

3.2.7 Safety valve. The system shall include a pressure relief valve to ensure against excessive pressure which may induce a personnel hazard.

3.2.8 Pressure gage. Pressure gages shall monitor both the effluent air pressure and the internal system pressure.

3.2.9 Controls. Controls shall include a manual circuit breaker type on-off switch and the control knob for the pressure regulator. Gage and controls shall be weather protected if the unit is to be pole mounted (see 6.2).

3.2.10 Materials. All air and water lines shall be constructed of corrosion resisting material.

3.2.11 Humidity alarm. A humidity alarm shall be provided to monitor the effluent air. If the moisture content of the dry air rises above the value specified the relay will close to give a signal (to the connected alarm system).

3.2.12 Interchangeability. All parts having the same manufacturer's part number shall be interchangeable.

3.2.13 Electrical. The dehydrator shall operate on 115, 208, 220, or 230 volt, single phase, 60 hertz power as specified in 6.2. All components and wiring shall comply with requirements of the National Electrical Code.

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3.2.14 Air inlet filter. An air inlet filter shall be provided to prevent insects, dust, or other foreign materials from entering the system.

3.2.15 Finishes and protective coatings.

3.2.15.1 Cleaning, painting, plating, anodic films, and chemical treatments. All cleaning, painting, plating, chemical films, and chemical treatment shall be in accordance with MIL-STD-808, Type II.

3.3 Workmanship. The dryer shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, sharp edges and accuracy of dimensions.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the air dryer shall be classified as follows:

a. Preproduction tests - see 4.3.
b. Acceptance tests - see 4.4.

4.3 Preproduction tests. One air dryer representative of the finished product in all respects shall be subjected to the tests specified in 4.5.

4.4 Individual tests. Each unit to be offered for acceptance shall be subjected to the tests of 4.5.1 and 4.5.2.2.

4.5 Test method.

4.5.1 Examination of product. The air dryer shall be inspected to ensure compliance with the requirements of this specification.

4.5.2 Operating test. The tests specified in 4.5.2.1 shall be conducted to verify satisfactory performance. Dew point and flow rate measurements of influent and effluent air shall be recorded at intervals which are adequate to determine performance.

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4.5.2.1 Endurance test. The air dryer shall be operated at ambient air conditions of 95° F dry bulb and 80° F wet bulb or at ambient specified in 6.2g. The unit shall be operated continuously at this condition for a time period of 500 hours. Throughout this test the effluent air dewpoint shall not be higher than specified in 1.2. The air dryer shall deliver the volume specified in 6.2 at the design pressure as specified in 6.2.

4.5.2.2 Individual test. Each unit shall be operated for a period of at least 8 hours before shipment. Throughout this test the effluent air, volume, pressure, and dew point shall be as specified in 6.2.

4.5.2.3 Humidity alarm. Operation of the humidity alarm shall be demonstrated by increasing the R. H. sufficiently to activate the alarm system.

4.5.3 Maintenance. Not more than one man-hour shall be consumed for maintenance during all of the preproduction test.

4.5.4 Rejection and retest. When an air dryer fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and corrected.

4.7 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully correct all defects.

4.8 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING.

5.1 General. The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Levels of preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1 Level A. Air dryers shall be preserved in accordance with MIL-P-116, Method II. Unit quantities shall be one each.

5.2.2 Level C. Air dryers shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.3 Levels of packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A. Air dryers preserved and packaged to meet 5.2.1 shall be packed in exterior type shipping containers meeting PPP-B-601. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.3.2 Level B. Air dryers preserved and packaged to meet 5.2.1 shall be packed in domestic type exterior containers meeting PPP-B-601. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with applicable container specification or appendix thereto.

5.3.3 Level C. Packages which require over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will ensure safe transportation at the lowest rate to the

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point of delivery. Containers shall meet consolidated freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.4 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

AIR DRYER

In addition, the following marking information shall be included:

LABEL, "HANDLE WITH CARE"

AND IDENTIFY "UP" POSITION

6 NOTES.

(This section contains information of a general or explanatory nature than may be helpful, but is not mandatory.)

6.1 Intended use. To provide low pressure dry air for the purpose of pressurizing cable systems or radar waveguide systems.

6.1.1 Suggestion for selection. The type II dryer will produce a lower dew point and is best suited where frost cannot be tolerated inside a pressurized system if the ambient temperature drops below the dew point. Frost inside of a waveguide will lower the quality of the signal and cause arcing. Therefore, a radar or microwave system located in extremely cold climates with prolonged temperatures of less that - 15° F should have a type II air dryer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type air dryer required (see 1.2.1).
c. Capacity (SCFD) required (see 1.2.2).
d. Pressure (psig) required (see 3.2.6).
e. Size limitations (height, length, width) (see 3.2.1).
f. When a weatherproof cabinet suitable for pole mounting is required (see 3.2.1).
g. Endurance test ambient condition, if different (see 4.5.2.1).
h. Operating voltage required (see 3.2.13).
i. Level of preservation, packaging and packing required (see 5.2 and 5.3).

6.3 Subject term (Key Word) listing.

Desiccant
Dryer
Effluent

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-83934B (USAF)	Dryer, Air, Low Pressure

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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